

Jared Siegel

Princeton University, Department of Astrophysical Sciences — Princeton, NJ 08544

✉ siegeljc@princeton.edu

🌐 jaredcsiegel.github.io

🆔 0000-0002-9337-0902

Education

Princeton University	Expected 2027
PhD in Astrophysics	
University of Chicago	2022
BA in Physics	
BS in Astrophysics	

Awards and Grants

Harold W. Dodds Fellowship <i>Princeton University</i>	2026–2027
Graduate Research Fellowship <i>Jane Street</i>	2026–2027
NSF Graduate Research Fellowship <i>National Science Foundation</i>	2022–2026
Centennial Fellowship <i>Princeton University</i>	2022–2027
Chambliss Astronomy Student Award <i>American Astronomical Society</i>	2020

Observing Proposals

PI James Webb Space Telescope Cycle 4 GO #8607	2025
“Studying cosmic noon at 200 parsec scales: resolved spectroscopy of a magnified dusty quiescent galaxy”	

Peer Reviewed Publications

- Cosmology and Galaxies
5. **Siegel, J.**, et al., *The suppression of the matter power spectrum: strong feedback from X-ray gas mass fractions, kSZ effect profiles, and galaxy-galaxy lensing*, submitted to The Monthly Notices of the Royal Astronomical Society, [arXiv:2512.02954](#)
 4. **Siegel, J.**, Amon A., et al., *Joint X-ray, kinetic Sunyaev-Zeldovich, and weak lensing measurements: toward a consensus picture of efficient gas expulsion from groups and clusters*, submitted to The Astrophysical Journal, [arXiv:2509.10455](#)
 3. **Siegel, J.**, McCullough, J., Amon A., et al., *Intrinsic alignment demographics for next-generation lensing: Revealing galaxy property trends with DESI Y1 direct measurements*, submitted to The Monthly Notices of the Royal Astronomical Society, [arXiv:2507.11530](#)

2. **Siegel, J.** & Melchior, P., *Spatially Resolved Galaxy-Dust Modeling with Coupled Data-Driven Priors*, The Astrophysical Journal, 986, 212 (2025), [arXiv:2411.08111](#)
1. **Siegel, J.**, Setton, D., Greene, J., et al., *UNCOVER: Significant Reddening in Cosmic Noon Quiescent Galaxies*, The Astrophysical Journal, 985, 125 (2025), [arXiv:2409.11457](#)

Extrasolar Planets

5. **Siegel, J.**, Halverson, S., Luhn, J., Zhao, L., Al Moulla, K., et al., *Quiet Please: Detrending Radial Velocity Variations from Stellar Activity with a Physically Motivated Spot Model*, The Astronomical Journal, 168, 158 (2024), [arXiv:2408.07121](#)
4. **Siegel, J.**, Winn, J., Albrecht, S., *Ponderings on the Possible Preponderance of Perpendicular Planets*, The Astrophysical Journal Letters, 950, 1 (2023), [arXiv:2305.13400](#)
3. **Siegel, J.** & Rogers, L., *Mass Upper Bounds for Over 50 Kepler Planets Using Low-S/N Transit Timing Variations*, The Astronomical Journal, 164, 139 (2022), [arXiv:2208.14398](#)
2. **Siegel, J.**, Rubenzahl, R., Halverson, S., & Howard, A., *Into the Depths: a new activity metric for high-precision radial velocity measurements based on line depth variations*, The Astronomical Journal, 163, 260 (2022), [arXiv:2204.05810](#)
1. **Siegel J.** & Fabrycky, D., *Resonant Chains of Exoplanets: Libration Centers for Laplace Angles*, The Astronomical Journal, 161, 290 (2021), [arXiv:2104.14665](#)

High Energy Astrophysics

3. **Siegel, J.**, Kiato, I., Kalogera, V., Berry, C., Maccarone, T., et al., *Investigating the Lower Mass Gap with Low Mass X-ray Binary Population Synthesis*, The Astrophysical Journal, 952, 212 (2023), [arXiv:2209.06844](#)
2. **Siegel, J.**, Dwarkadas, V. V., Frank, K. A., & Burrows, D. N., *Can the Fe K-alpha line reliably predict supernova remnant progenitors?*, The Astrophysical Journal, 922, 67 (2021), [arXiv:2109.01157](#)
1. **Siegel, J.**, Dwarkadas, V. V., Frank, K., & Burrows, D. N., *Analysis of XMM-Newton Observations of Supernova Remnant W49B and Clues to the Progenitor*, The Astrophysical Journal, 904, 175 (2020), [arXiv:2010.04765](#)

Contributing Author

2. Bigwood L., McCullough J., **Siegel, J.**, et al., *Confronting cosmic shear astrophysical uncertainties: DES Year 3 revisited*, submitted to PRD, [arXiv:2512.04209](#)
1. Bigwood L., Yamamoto M., **Siegel, J.**, et al., *The kinetic Sunyaev Zeldovich effect as a benchmark for AGN feedback models in hydrodynamical simulations: insights from DESI + ACT*, submitted to The Monthly Notices of the Royal Astronomical Society, [arXiv:2510.15822](#)

Selected Talks

Seminar [<i>Invited</i>]	Harvard University, Apr. 2026
CMB/LSS Seminar	Cambridge University, Apr. 2026
Physics Seminar [<i>Invited</i>]	University of Alabama, Apr. 2026
Cosmology Seminar [<i>Invited</i>]	UC Berkeley, Mar. 2026
AGN & Energy Flows Workshop	Center For Computational Astrophysics, Feb. 2026
Mock NYC Workshop	Center For Computational Astrophysics, Jan. 2026
Cosmology Seminar [<i>Invited</i>]	Yale University, Nov. 2025
European Southern Observatory MMC2025	Italy, Oct. 2025
Cosmic Ecosystems Conference	Perimeter Institute, Canada, Jul. 2025
Millimeter Universe Conference	University of Chicago, Jun. 2025
Big Data, Big Questions: The Future of Cosmological Surveys	Germany, May 2025

Outreach and Service

Princeton Prison Teaching Initiative	Spring 2025—Present
Developed curricula, served as an in-class tutor, and oversaw the teaching team for AST 207 “Gateway to Science” at East Jersey State Prison.	

Teaching

Teaching assistant—Princeton University	
AST 401 Cosmology	Spring 2026
APC 524 Software Engineering for Scientific Computing	Fall 2025
AST 207 Gateway to Science: Observational Astronomy	Spring 2025
AST 205 Planets in the Universe	Fall 2023
Teaching assistant—University of Chicago	
ASTR 211 Computational Techniques in Astrophysics	Spring 2021
ASTR 205 Introduction to Python Programming	Fall 2020